to copy content) will fail. This is because the CD will appear to hold data that exceeds its capacity and the OS will refuse to copy the files. The application that uses the content on the CD has pre-knowledge of the nature of the copy-protection scheme used. It thus knows the true length of the files involved and the CD works as intended. This was one of the first methods used and is no longer effective, thanks to the prevalence of 80-minute CD-R media and the arrival of software that lets you overburn a CD—that is, attempt to write more content on a CD than what it can actually store. Other protection techniques create an illegal Table of Contents (TOC) to similar effect, i.e., making a disc appear larger than it actually is. (Every CD-ROM has an indexed listing of its contents that forms the TOC). Other methods use a non-standard means of storing data on the CD. Data is stored on a CD in the form of tracks. To protect Audio CDs, atypical gaps between audio tracks or the use of tracks with an uncommon length of less than 4 seconds are used.

Introduction of errors: Some CD-ROMs are physically damaged by a line or a ring across the data track to protect their content. Most CD-Writers are not able to copy data from a medium which has this kind of an error. Other protection schemes introduce errors while mastering a CD-ROM disc. The application that is meant to use the CD, checks for the anomaly and will accept the CD as a legitimate version only if the introduced errors are present. If someone tries to copy the CD, the writing software will use its error-correction facility to correct the inaccuracies that were introduced during the mastering process. The copy will then be faultless, but the application will fail to detect the error and will thus refuse to run. This form of protection is generally used in CD-ROMs containing non-audio data content.

A somewhat fancier method of introducing a protection scheme on an Audio CD involves changing the ECC (Error Correction Code) information. ECC allows data that is being read or transmitted to and from a CD-ROM to be checked for errors and, when necessary, corrected on the fly. It differs from parity-checking in that errors are not only detected but also corrected. To guard the content of an Audio CD, the ECC data present is scrambled to appear as a non-correctable error. When an Audio CD is played on a computer system, the data is transported from the CD-ROM drive to the soundcard via an analogue path. Along this path are logic elements that are designed to detect any errors and use software techniques to smoothen out the flawed audio. When an audio track is copied through a CD-Writer however, the data takes a digital path meant for generic non-audio data. The error correction is absent along this route and thus the original error that was introduced remains. This error comes across as scratching and hissing noise in the copied audio track. Newer writers use data correction at lower speeds and this protection method can be thus bypassed.

Encryption: Encryption technology has been successfully used as a means to safeguard content. The main executable files on a protected CD-ROM are first encrypted using a special digital signature. The signature is then mapped to a software decryption key. The decryption will thus only happen if the signature is present. The required protection is offered by the fact that the digital signature is not reproducible by either counterfeiting (re-mastering) or disc burning. When an authentic disc is used, the signature is present, decryption occurs and the application runs. A pirate who attempts to replicate such a disc will only be able to copy its content and not the embedded signature. When a copy is thus used, the signature is not present, no decryption occurs and the application does not run.

Inherent protection: Every CD has important information on the disc itself which is called ATIP (Absolute Time In Pre-groove, or pre-groove for short). The ATIP contains information that indicates the

> “We’re working very hard to make buying music as easy as stealing music. And we’re also working hard to make stealing a hell of a lot harder”
>
> *— Jay Samit, EMI’s senior vice president*

Foiled again!

Code Breaker

A software to decrypt content scrambling system (CSS), called DeCSS, was released in October 1999. The DeCSS information can be used to guess the master keys that form the base of CSS protection scheme, a move which is considered illegal under the Digital Millennium Copyright Act (DMCA). On January 14, 2000, the seven top US movie studios—Disney, MGM, Paramount, Sony, Time Warner, Twentieth Century Fox and Universal—backed by the Motion Picture Association of America (MPAA), filed lawsuits in an attempt to stop the distribution of DeCSS on Web sites. On January 24, 2000, 16-year-old Jon Johansen, the Norwegian programmer who first distributed DeCSS, was questioned by the local police who raided his house. Johansen said that the actual cracking work was done by two anonymous European programmers, who called themselves Masters of Reverse Engineering (MoRE). But the cat was already out of the bag as the DeCSS code was made public during the court proceedings. It can now be found on the back of T-shirts everywhere!

Marked as Defeated

A copy-protected Celine Dion album offended music lovers around the world as it hung PCs and crashed some iMacs so hard that they had to be serviced by dealers. Use of a mixed-mode disc (containing both audio and data) was identified as the cause of the problem. By simply masking the second Table of Contents on the disc (which appears as a thin ring on the recorded surface, about half an inch from the outer edge), music-buffs bypassed the protection scheme. The masking was done via decidedly low-tech means—by using a black marker pen!